

Section B

Q4b) Control parameter K_r :

(11)

$$C(A-BK)^{-1}B = (0 \ 1) \begin{pmatrix} 1-K_1 & -1-K_2 \\ -2 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\frac{1}{(1-K_1)(2) - (-1-K_2)(-2)} \begin{pmatrix} 2 & 1+K_2 \\ 2 & 1-K_1 \end{pmatrix}$$

$$= \frac{1}{2-2K_1-2-2K_2} \begin{pmatrix} 2 & 1+K_2 \\ 2 & 1-K_1 \end{pmatrix}$$

$$= \frac{1}{-2K_1-2K_2} \begin{pmatrix} 2 & 1+K_2 \\ 2 & 1-K_1 \end{pmatrix}$$

$$C(A-BK)^{-1}B = \frac{1}{-2K_1-2K_2} (0 \ 1) \begin{pmatrix} 2 & 1+K_2 \\ 2 & 1-K_1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \frac{1}{-2K_1-2K_2} (2 \ 1-K_1) \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \frac{1}{-2K_1-2K_2} (2)$$

$$= \frac{2}{-2K_1-2K_2} = \frac{1}{-K_1-K_2}$$

$$U = -K_1 C_1 - K_2 C_2 + 2K_r$$

$$U = -K_1 C_1 - K_2 C_2 + 2 \left(\frac{1}{-K_1-K_2} \right)$$

$$U = -K_1 C_1 - K_2 C_2 + \frac{2}{-K_1-K_2}$$

$$\text{where } K_1 = 5 \text{ and } K_2 = -5.5$$

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Q4b)
cont.

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$$U = -(K_1 \ K_2) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + 2K_r r$$

$$U = -K_1 c_1 - K_2 c_2 + 2K_r r$$

Closed loop system becomes

$$\frac{dc}{dt} = (A - BK)c + BK_r r$$

$$BK = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (K_1 \ K_2) = \begin{pmatrix} K_1 & K_2 \\ 0 & 0 \end{pmatrix}$$

$$A - BK = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} - \begin{pmatrix} K_1 & K_2 \\ 0 & 0 \end{pmatrix} \\ = \begin{pmatrix} 1-K_1 & -1-K_2 \\ -2 & 2 \end{pmatrix}$$

$$\frac{dc}{dt} = \begin{pmatrix} 1-K_1 & -1-K_2 \\ -2 & 2 \end{pmatrix} c + 2 \begin{pmatrix} K_r \\ 0 \end{pmatrix}$$

$$y = Cc = c_2$$

$$\det(A - BK - \lambda) = \begin{vmatrix} 1-K_1-\lambda & -1-K_2 \\ -2 & 2-\lambda \end{vmatrix}$$

$$= (1-K_1-\lambda)(2-\lambda) - (-1-K_2)(-2)$$

$$= 2 - \lambda - 2K_1 + \lambda K_1 - 2\lambda + \lambda^2 - (2 + 2K_2)$$

$$= \lambda^2 - 3\lambda + \lambda K_1 + 2 - 2K_1 - 2 - 2K_2$$

$$= \lambda^2 - 3\lambda + \lambda K_1 - 2K_1 - 2K_2$$

$$= \lambda^2 + (-3+K_1)\lambda - 2K_1 - 2K_2$$

$$\text{Eigenvalues} = \lambda_1 = -1, \lambda_2 = -1$$

$$(\lambda+1)(\lambda+1) = \lambda^2 + \lambda + \lambda + 1$$

$$= \lambda^2 + 2\lambda + 1$$

$$\lambda^2 + (-3+K_1)\lambda - 2K_1 - 2K_2 = \lambda^2 + 2\lambda + 1$$

$$-3+K_1 = 2$$

$$K_1 = 5$$

$$-2K_1 - 2K_2 = 1$$

$$-2(5) - 2K_2 = 1$$

$$-2K_2 = 11$$

$$K_2 = -5.5$$

Section B

⑨

$$Q4) \begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u, \quad y = c_2$$

$$y = (0 \ 1) c_2$$

$$Q4a) A = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad C = (0 \ 1)$$

Observable

$$W_o = \begin{pmatrix} C \\ CA \end{pmatrix}$$

$$CA = (0 \ 1) \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} = (-2 \ 2)$$

$$W_o = \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}$$

$\therefore$ the system is observable as the rows are linearly independent.

Reachable

$$W_r = (B \ AB)$$

$$AB = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$W_r = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} \quad \det(W_r) = -2$$

$\therefore$ the system is reachable as the $\det \neq 0$.

$$Q4b) u = -Kc + K_r r \quad K = (K_1 \ K_2) \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$y = r = 2 \quad \lambda = -1, \lambda = -1$$

$$\frac{dc}{dt} = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u, \quad y = (0 \ 1) c_2$$

$$A = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad C = (0 \ 1)$$

$$W_r = (B \ AB) = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}, \quad W_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}$$

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Q2c) longer than 3

$$\alpha = 300 \text{ hour}^{-1} = 5 \text{ min}^{-1}$$

$$\beta = 4 \text{ min}^{-1}$$

$$\pi_k = \frac{\rho^k e^{-\rho}}{k!}$$

$$\rho = \frac{5}{4}$$

$$k = 3$$

$$\pi_B = \frac{\left(\frac{5}{4}\right)^3 e^{-\frac{5}{4}}}{3!}$$

$$\pi_3 = 0.0932$$

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Q2a) ii)
cont.

$$\Rightarrow \pi_0 = \frac{1}{\sum_{k=0}^r \rho^k}$$

$$\Rightarrow \pi_0 = \frac{1}{\frac{1-\rho^{r+1}}{1-\rho}} = \frac{1-\rho}{1-\rho^{r+1}}, \rho \neq 1$$

$$\pi_0 = \frac{1}{r+1}, \rho = 1$$

Q2b) $\alpha_k = \alpha$, $\beta_k = k\beta$, $\alpha > 0$, $\beta > 0$

Equilibrium probabilities π_k , $k = 0, 1, 2, \dots$

$$\pi_k = \frac{\rho^k e^{-\rho}}{k!}, \rho = \frac{\alpha}{\beta}$$

$$\pi_k = \frac{\rho^k}{k!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \dots + \frac{\rho^r}{r!}} \quad \rho = \frac{\alpha}{\beta}$$

$$\Rightarrow \pi_k = \frac{\rho^k}{k!} e^{-\rho} \frac{1}{\sum_{k=0}^r \frac{\rho^k}{k!}} \quad k = 0, 1, 2, \dots, r$$

$$\pi_0 = \frac{1}{\sum_{k=0}^r \frac{\rho^k}{k!}}$$

Since $\sum_{k=0}^r \pi_k = 1$ we get $\pi_0 \sum_{k=0}^r \frac{\rho^k}{k!} = 1$

$$\pi_k = \frac{\rho^k e^{-\rho}}{k!} \quad \text{where } \rho = \frac{\alpha}{\beta}$$

Section B

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Q2 a) i) max queue size = r

The probability of arrival and servicing is independent of the queue size. Only one customer can arrive or be serviced in one time interval hence: $\alpha_r = \beta_0 = 0$

Q2 a) ii) Shown in two steps

$$(1) \pi_k = p \pi_{k-1} \Rightarrow \pi_k = p^2 \pi_{k-2} = \dots = p^k \pi_0$$

$$(2) \pi_0 = \frac{1-p}{1-p^{r+1}}, \quad p \neq 1, \quad \pi_0 = \frac{1}{1+r}, \quad p = 1$$

(i) by induction:

using the transition matrix P we have

$$\pi = \pi P, \quad \pi = (\pi_0, \dots, \pi_r)$$

$$\pi_0 = \pi_0(1-p_1) + \pi_1 p_2$$

$$0 = -\pi_0 p_1 + \pi_1 p_2$$

$$\pi_1 p_2 = \pi_0 p_1$$

$$\pi_1 = \frac{p_1}{p_2} \pi_0 = p \pi_0$$

(i) holds for $k=1$

(2) Assume $\pi_{k-1} = p \pi_{k-2}$ holds then we have $(k-1)^{\text{st}}$ element of $\pi = \pi P$ gives

$$\pi_{k-1} = \pi_{k-2} p_1 + \pi_{k+1} (1-p_1-p_2) + \pi_k p_2$$

$$\Rightarrow p_2 \pi_k = -\pi_{k-2} p_1 + (p_1 + p_2) \pi_{k-1}$$

$$\Rightarrow \{ \pi_{k-1} = p \pi_{k-2} \Rightarrow p_2 \pi_{k-1} = p_1 \pi_{k-2} \}$$

$$p_2 \pi_k = \frac{-p_2 \pi_{k-1}}{p_2} + (p_1 + p_2) \pi_{k-1}$$

$$= \pi_k = \frac{p_1}{p_2} \pi_{k-1} = p \pi_{k-1}$$

so (i) holds for k

$$\Rightarrow \pi_k = p^k \pi_0$$

We know $\sum_{k=0}^r \pi_k = 1$

$$\Rightarrow \sum_{k=0}^r p^k \pi_0 = 1$$

Section A

Q1c) Sub $P_1 = 20 - P_2$ into $\dot{P}_2 = 0$
cont.

$$3P_2(10 - 20 + P_2 - P_2) = 0$$

$$3P_2(-10) = 0$$

$$P_2 = 0 \quad 3(0)(-10) = 0$$

$$P_1 = 20 - 0 = 20$$

$$\text{Equilibrium Point} = (20, 0)$$

Q1d) $\frac{dx}{dt} = \begin{pmatrix} 0 & b \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$

$$y = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & b \\ 1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Q1d) i) $W_o = \begin{pmatrix} C \\ CA \end{pmatrix}$

$$CA = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & b \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix}$$

$$W_o = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{The rows are linearly independent.} \\ \therefore \text{the system is observable}$$

Q1d) ii) $W_r = \begin{pmatrix} B & AB \end{pmatrix}$

$$AB = \begin{pmatrix} 0 & b \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ 1 \end{pmatrix}$$

$$W_r = \begin{pmatrix} 0 & b \\ 1 & 1 \end{pmatrix} \quad \det(W_r) = -b$$

$\therefore$ the system is reachable if $b \neq 0$

Section A

④

$$Q1c) i) \frac{dP_1}{dt} = 40P_1 - 2P_1^2 - 2P_1P_2$$

$$\frac{dP_1}{dt} = 2P_1(20 - P_1 - P_2)$$

$$\frac{dP_2}{dt} = 30P_2 - 3P_1P_2 - 3P_2^2$$

$$\frac{dP_2}{dt} = 3P_2(10 - P_1 - P_2)$$

P_1 = population size of P_1

P_2 = population size of P_2

20 = maximum population of P_1

10 = maximum population of P_2

$2P_1$ = biotic population for P_1

$3P_2$ = biotic population for P_2 .

$$Q1c) ii) \dot{P}_1 = 0, \dot{P}_2 = 0 \rightarrow \text{Equilibrium parts}$$

$$\dot{P}_1 = 0 \Rightarrow 2P_1(20 - P_1 - P_2) = 0$$

$$P_1 = 0 \quad 2(0)(20 - 0 - P_2) = 0$$

$$0 = 0$$

$$P_1 = 20 - P_2 \quad 2(20 - P_2)(20 - 20 + P_2 - P_2) = 0$$

$$2(20 - P_2)(0) = 0$$

$$0 = 0$$

$$\text{Sub } P_1 = 0 \text{ into } \dot{P}_2 = 0$$

$$3P_2(10 - P_1 - P_2) = 0$$

$$P_1 = 0 \quad 3P_2(10 - 0 - P_2) = 0$$

$$P_2 = 0 \quad 3(0)(10 - P_2) = 0$$

$$0 = 0$$

$$\text{Equilibrium point} = (0, 0)$$

$$P_2 = 10 \quad 3(10)(10 - 10) = 0$$

$$0 = 0$$

$$\text{Equilibrium point} = (0, 10)$$

b)

i)

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{pmatrix} p & 0.2 & 0 \\ 0 & q & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix} \end{matrix}$$

$$p + 0.2 = 1$$

$$p = 0.8$$

$$q + 0.3 = 1$$

$$q = 0.7$$

Q1b)

ii) Yes the process is ergodic as you can travel to each states using the other states. i.e. there is a connection between the states where you can visit each state.

Q1b iii) Yes the process has unique equilibrium probabilities.

$$(\pi_1 \ \pi_2 \ \pi_3) \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$0.8\pi_1 + 0.2\pi_2 = \pi_1$$

$$0.7\pi_2 + 0.3\pi_3 = \pi_2$$

$$0.5\pi_1 + 0.5\pi_3 = \pi_3$$

$$0.2\pi_2 = 0.2\pi_1$$

$$\pi_2 = \pi_1$$

$$0.3\pi_2 = 0.3\pi_3$$

$$\pi_2 = \pi_3$$

$$\pi_2 (1 + 1 + 1) = 1$$

$$3\pi_2 = 1$$

$$\pi_2 = \frac{1}{3}$$

$$\pi_1 = \frac{1}{3}$$

$$\pi_3 = \frac{1}{3}$$

Equilibrium probabilities =

$$\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

Section A

Q1 a)

$$f(t) = \begin{cases} \frac{2t+1}{4}, & 1 \leq t \leq 2 \\ 0, & 0 \leq t < 1, t > 2 \end{cases}$$

i) $\int_{-\infty}^{\infty} f(t) dt = 1$ = probability density function

$$\int_1^2 \frac{1}{2}t + 1 = \left. \frac{1}{2} \frac{t^2}{2} + t \right|_1^2 = \frac{3}{4}$$

ii) $\sum_{1 \leq t \leq b} \frac{2t+1}{4}$

ii) Variance = σ^2 Standard deviation = σ

$$\sigma^2 = \langle t^2 \rangle - \mu^2$$

$$\mu = \langle t \rangle = \sum_{t \geq 0} t p_t$$

$$= \sum_{1 \leq t \leq b} t \left(\frac{2t+1}{4} \right)$$

$$\sigma^2 = \sum_{1 \leq t \leq b} t \left(\frac{2t+1}{4} \right) - \sum_{1 \leq t \leq b} \frac{(2t+1)^2}{16} = \text{Variance}$$

~~$$\sigma^2 = \sum_{1 \leq t \leq b} t \left(\frac{2t+1}{4} \right) - \sum_{1 \leq t \leq b} \frac{(2t+1)^2}{16} = \text{Variance}$$~~

$$\sigma = \sqrt{\sum_{1 \leq t \leq b} t \left(\frac{2t+1}{4} \right) - \sum_{1 \leq t \leq b} \frac{(2t+1)^2}{16}} = \text{Standard deviation}$$